System and Devices (SYS 3)

Tutorial Lecture 4 : Cache Memory

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COM00018I

Cache Block Concepts

For a 32KB direct mapped cache with 64 byte cache block, give the address of the starting byte of the first word in the block that contains the address 0X7245E824.

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32KB direct mapped cache with 64 byte cache block address 0X 7245E824.

sets = CS/(BSXA) = $2^{15}/(2^{6}X1) = 2^{9}$ = 512 sets

32KB= 2^{5} 2^{10} = 2^{15}

Tag	Index = 9	Offset = 6

Address: 0X 7245E824

-----------0010 0100 offset (6 bits)

Block address range from -----0000 0000 to ----- 0011 1111

0X 7245E800 to 0X 7245E83F

Hence, starting address if 0X 7245E800.

Index and Offset Calculations

A cache has 512 KB capacity, 4B word, 64B block size and 8-way set associative. The system is using 32 bit address. Given the following addresses, which set of cache will be searched and specify which word of the selected cache block will be forwarded if it is a hit in cache?

(a) 0X ABC89984 (b) 0X 485669AC

Index and Offset Calculations

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(a) 0X ABC89984   (b) 0X 485669AC

sets = CS/(BSXA) = $2^{19}/(2^{6}X2^{3}) = 2^{10}$ = 1024 sets

1 word = 4B, Hence 64 byte block has 16 words.

Tag = 16	Index = 10	Offset = 6 (4+2)

0X ABC89984 = 1001 1001 1000 0100 → Set 614, word 1

0X 485669AC = 0110 1001 1010 1100 → Set 422, word 11

Tag and Data Array Access

It takes 3 ns to access a tag array value and 4.2 ns to access a data array, 1.3 ns to perform hit/miss comparison and 1 ns to return the selected data to processor. Answer the following questions:

(1) What is the cache hit latency of the system?

(2) What is the cache hit latency of the system if hit/miss comparison will take 0.6 ns?

(3) What would be the hit latency if both tag and data array access take 3.5 ns and hit/miss comparison take 1 ns?

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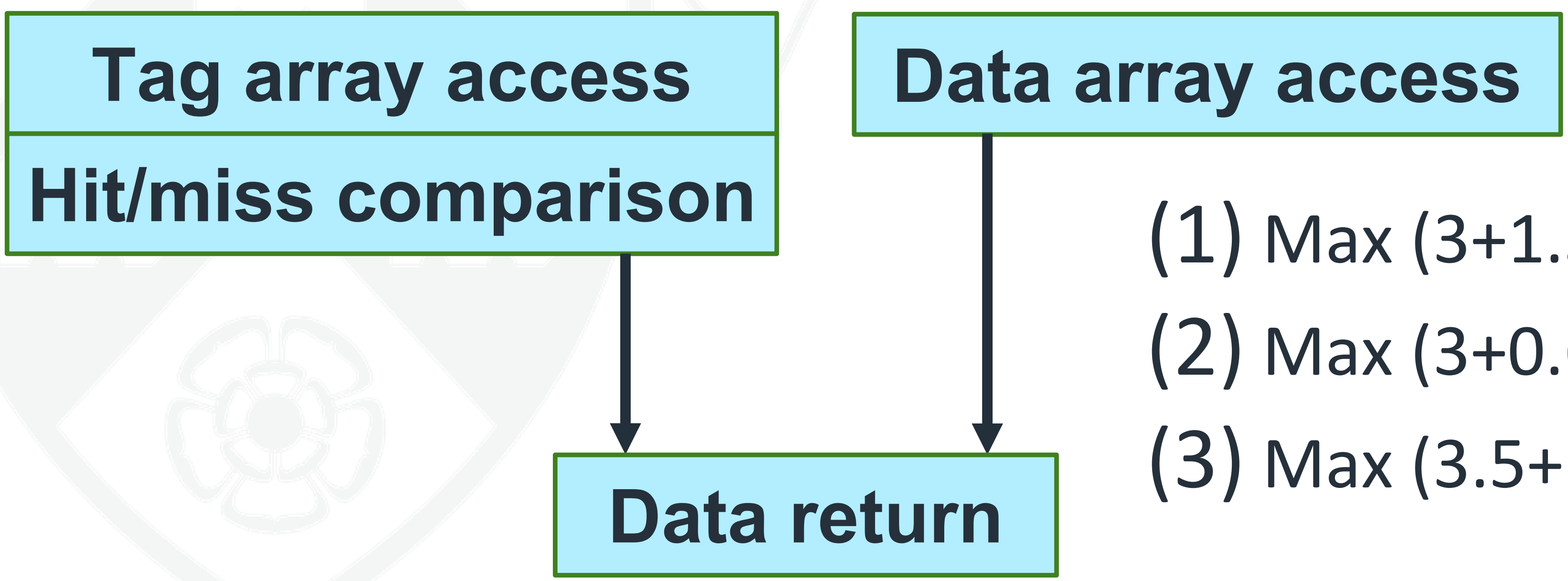

(1) Max (3+1.3, 4.2) +1 = 5.3ns

(2) Max (3+0.6, 4.2) +1 = 5.2ns

(3) Max (3.5+1, 3.5) +1 = 5.5ns

Multilevel Caches

Assume a 2-level cache system with the following specifications. L1 Hit Time = 1 cycle, L1 Miss Rate = 2.5%, L2 Hit Time = 6 cycles, L2 Miss Rate = 17% (% L1 misses that miss), L2 Miss Penalty = 120 cycles. Compute the average memory access time.

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Average memory access time = Hit time + (Miss rate × Miss penalty)

AMAT = ht1 + mr1 x MP

= ht1 + mr1 x (ht2 + mr2 x MP)

= 1 + 0.025 x (6 + 0.17 x 120) = 1.66 CC

Summary